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Abstract

Measures of self-esteem frequently conflate two independent constructs: high self-esteem (a normative positive sense of self) and narcissistic grandiosity (a nonnormative sense of superiority). Confusion stems from the inability of self-report self-esteem scales to adequately distinguish between high self-esteem and narcissistic grandiosity. The Narcissistic Grandiosity Scale (NGS) was developed to clarify this distinction by providing a measure of narcissistic grandiosity. In this research, we refined the NGS and demonstrated that NGS scores exhibit good convergent, discriminant, and concurrent validity relative to scores on theoretically relevant measures. NGS scores, when used as simultaneous predictors with scores on a self-esteem measure, related more strongly to phenomena linked to narcissistic grandiosity (e.g., competitiveness, overestimating one's attractiveness, lack of shame), whereas self-esteem scores related more strongly to phenomena crucial to individuals' well-being (e.g., higher levels of optimism and satisfaction with life, and lower levels of depression, worthlessness, and hostility). The NGS provides researchers with a measure to help clarify the distinctions between narcissistic grandiosity and high self-esteem, as well as other facets of narcissism, both in theory and as predictors of important real-life characteristics.

Keywords

narcissism, narcissistic grandiosity, grandiose narcissism, narcissistic personality disorder, Narcissistic Grandiosity Scale, self-esteem, discriminant validity

Can a person's self-esteem be too high? According to sociologist Morris Rosenberg (1965), in his seminal book on self-esteem, the answer to this question is "no." It is not that people cannot hold overly positive self-views. They can. However, Rosenberg considers self-views that are *too* positive as falling outside the boundaries of self-esteem altogether. As Rosenberg described it, high self-esteem is "feeling that one is 'good enough' . . . [not] the ultimate in perfection but, on the contrary, [recognizing one's] limitations" (p. 31). Thus, from Rosenberg's perspective, an inflated, superior sense of oneself is not a variant of high self-esteem. Rather, an inflated sense of self belongs to an independent, differentiable construct (for similar views, see Brummelman, Thomaes, & Sedikides, 2016; Crowe, Sleep, Carter, Campbell, & Miller, 2018; Rosenthal & Hooley, 2010; Tracy, Cheng, Robins, & Trzesniewski, 2009).

Unfortunately, the highly utilized self-esteem scale Rosenberg (1965) developed does not do a good job differentiating "good enough" high self-esteem from an inflated positive self-view outside the boundaries of high self-esteem (e.g., Jordan, Spencer, Zanna, Hoshino-Browne, & Correll, 2003; Tafarodi & Ho, 2003). For example, high scores on the Rosenberg Self-Esteem Scale (RSES) cannot distinguish between a self-confident respondent who has a normatively positive self-view and an insufferably egotistical respondent who has an unrealistically inflated sense of self (see Coopersmith, 1959). Both are likely to provide equally high ratings on RSES items such as "On the whole, I am satisfied with myself." However, the latter's high RSES score comes not from having high self-esteem, but rather from holding high levels of a different trait. That trait is narcissistic grandiosity. The goal of this research is to summarize the current state of the science on a scale we developed to measure narcissistic grandiosity—the Narcissistic

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Grandiosity Scale (NGS; see Rosenthal, 2005)—and to optimize it to better discriminate between high self-esteem and narcissistic grandiosity.

Narcissistic grandiosity is an individual's sense of self-importance and superiority in ability and/or status (e.g., Brummelman et al., 2016; Tracy et al., 2009). Grandiosity is regarded as a central feature of narcissism, both from a social/personality and a clinical perspective (Ackerman, Hands, Donnellan, Hopwood, & Witt, 2017). Although not necessarily pathological, narcissistic grandiosity is also not the same thing as "extremely high" self-esteem (see Brummelman et al. for a review).[1] For example, narcissistic grandiosity is related to feeling superior to others, whereas high self-esteem is not (Horvath & Morf, 2010). Behavior-wise, narcissistic grandiosity is related to a desire to dominate others rather than to bond with them (Campbell, Rudich, & Sedikides, 2002)—it is related to getting ahead, whereas high self-esteem is related to getting along (Brummelman et al.). Individuals with high levels of narcissistic grandiosity crave constant validation from others (Morf & Rhodewalt, 2001), whereas individuals with high self-esteem do not. Narcissistic grandiosity and self-esteem generally rise and fall at different periods over the life course, and they "arise from distinct forms of regard" from other people (Brummelman et al., 2016, p. 10)—narcissistic grandiosity is linked to overvaluation by parents whereas high self-esteem is fostered by parental warmth (Brummelman et al., 2015).

Measuring High Self-Esteem Versus Narcissistic Grandiosity

One particularly influential article (Baumeister, Campbell, Krueger, & Vohs, 2003) interpreted the heterogeneity of individuals with high scores on measures of self-esteem—that some have normative feelings of being "good enough" whereas others have an inflated, superior sense of self—as reflecting the nature of self-esteem itself rather than as a problem with the measurement of self-esteem: "[S]elf-esteem is literally defined by how much value people place on themselves…[it] does not carry any definitional requirement of accuracy . . . self-esteem is thus perception rather than reality" (p. 2). From this perspective, narcissistic grandiosity is not a different construct, but rather is "one subcategory of high self-esteem" (Baumeister et al., 2003, p. 24; see also Baumeister, Smart, & Boden, 1996) that lies at the top of the self-esteem continuum. As a result, Baumeister et al. proposed that self-esteem is not a particularly useful construct to study because high self-esteem scores do not predict many of the positive outcomes typically thought of as associated with high self-esteem. Furthermore, they concluded that there is no solution to this measurement challenge because "there is no objective criterion against which to compare self-reported self-esteem . . . [thus] there is no known basis for saying that certain people really have more or less self-esteem than they think they have" (Baumeister et al., 2003, p. 5).

Alternatively, numerous researchers have identified as problematic the inability of the RSES (and other self-esteem scales) to distinguish between "accurate" and "justified" self-assessments (i.e., high self-esteem) on the one hand, and "inflated" and "unwarranted" self-assessments (i.e., narcissistic grandiosity) on the other hand (e.g., Brummelman et al., 2016; Donnellan, Trzesniewski, Robins, Moffitt, & Caspi, 2005; Horvath & Morf, 2010; Myers & Zeigler-Hill, 2012; Orth, Robins, Meier, & Conger, 2016; Paulhus, Robins, Trzesniewski, & Tracy, 2004; Sedikides, Rudich, Gregg, Kumashiro, & Rusbult, 2004).[2] One frequent solution employed is to control for the effects of narcissism when conducting self-esteem research by partialing out the variance associated with a composite narcissism measure such as the Narcissistic Personality Inventory (NPI; Raskin & Hall, 1979, 1981). This step helps reveal the effects of "pure" self-esteem with the confounding effects of narcissism statistically removed. In fact, in reassessing their 2003 work, Baumeister and Vohs (2018) even hinted that they recognized that controlling for narcissism might help clarify self-esteem's role in various outcomes. Regardless, without a measure specific to narcissistic grandiosity, rather than a broad composite narcissism measure such as the NPI, such analyses unnecessarily control for features of narcissism that are less central to confounding the measurement of self-esteem, such as entitlement, exploitativeness, lack of empathy, and so on. Relying solely on a composite narcissism measure has the potential to produce less interpretable and meaningful analyses because there is less clarity regarding the reasons the narcissism covariate affected the results. For this reason, measures of specific facets of narcissism, including grandiosity, are desirable.

The Narcissistic Grandiosity Scale

The Narcissistic Grandiosity Scale (initially called the State-Trait Grandiosity Scale, see Rosenthal, 2005) was developed to address the lack of a measure of narcissistic grandiosity.[3] The NGS is a 16-item self-report adjective-rating scale of grandiose self-assessments modeled on the Positive and Negative Affect Scale (Watson, Clark, & Tellegen, 1988). It can be used for both trait (i.e., general) and state (i.e., current) self-reports. The initial scale validation data revealed that the potential NGS items at the extreme level of narcissistic grandiosity (e.g., "superior," "envied") did a better job of discriminating grandiosity from high self-esteem than did items representing more normative positive self-assessments (e.g., "confident," "attractive"). Additional details about the original NGS

item-selection are presented in the Supplementary Material (see Table S1 available online).

An unpublished version of the NGS was subsequently presented at academic conferences. In 2006, independent researchers began to publish data that included the NGS. As of mid-2018, the NGS had been cited in more than 50 peer-reviewed journal articles, 34 of which report NGS data. Independent published research indicates that NGS scores exhibit strong psychometric properties (e.g., weighted mean $\alpha = .95$),[4] good convergent validity in relation to grandiose narcissism (i.e., NPI) scores (meta-analytic $r = .53$), and in comparison, good discriminant validity from self-esteem (e.g., RSES) scores (meta-analytic $r = .25$). Additional details about this analysis are presented in the Supplementary Material (see Table S2 available online).

Independent research using the NGS reveals that a measure specific to narcissistic grandiosity is consequential for the field. First, NGS scores have a unique nomological and predictive network that is consistent with theoretical conceptions of narcissistic grandiosity and differs from the nomological network of scores on measures of self-esteem. For instance, NGS scores converge with clinical and expert-rated narcissistic personality disorder measures, which are either uncorrelated or negatively correlated with self-esteem (Crowe, Carter, Campbell, & Miller, 2016). NGS scores share relations with a range of personality traits reflecting the nomological network of a narcissism measure (e.g., low agreeableness, low honesty/humility), but not of a self-esteem measure (Crowe et al., 2016; Watts, Waldman, Smith, Poore, & Lilienfeld, 2017). NGS scores also share a relation with self-enhancing, unrealistic optimism that is independent of self-esteem (Tamborski, Brown, & Chowning, 2012). And when narcissistic grandiosity and self-esteem are used as simultaneous predictors, narcissistic grandiosity, but not self-esteem, is associated with psychopathic meanness, whereas self-esteem, but not narcissistic grandiosity, is (negatively) associated with psychopathic disinhibition (Donnellan & Burt, 2016). Results such as these prompted Crowe et al. (2016) to conclude that NGS scores are "not a simple proxy for positive self-evaluation" (p. 1556). Taken together, independent research using the NGS supports the view that although narcissistic grandiosity and self-esteem "both entail positive views of the self . . . these views are qualitatively different . . . [with] distinct phenotypes" (Brummelman et al., 2016, p. 9; see also Krizan & Herlache, 2017).

Second, research using the NGS indicates that it has different correlates from, and predicts different outcomes than, measures of other facets of grandiose narcissism (e.g., entitlement, exploitativeness). For example, grandiosity and entitlement function independently in relation to mental health and ethical misconduct (Brown, Budzek, & Tamborski, 2009) and in the use of self-serving self-enhancement strategies (Tamborski et al., 2012). Furthermore, grandiosity, entitlement, and exploitativeness each predict different aspects of health-risk behavior in young adults (Buelow & Brunell, 2014). Additionally, factor analysis reveals that the NGS and measures of entitlement and exploitativeness delineate separate traits (Brunell & Buelow, 2018). And a study comparing mono- and dizygotic twins indicates that narcissistic grandiosity and entitlement are independent heritable traits (Luo, Cai, & Song, 2014). In summary, the NGS helps differentiate narcissistic grandiosity from both self-esteem and other facets of grandiose narcissism and provides specific information that cannot be gleaned from a composite measure of narcissism such as the NPI.

Crowe et al. (2016). In addition to validation work on the NGS, Crowe et al. (2016) included analyses aimed at reducing the number of items in the scale. Their proposal for a 6-item version of the scale has subsequently been recommended for use by independent researchers (Foster et al., 2018). Unfortunately, the method Crowe et al. used for examining and recommending items—item function in an item response theory (IRT) analysis—was not ideal for creating a shortened version of the NGS.

One source of concern with the approach taken by Crowe et al. (2016) is that IRT assesses each scale item relative to the underlying construct operationalized by the scale itself (Steinberg & Thissen, 1995). Decisions to retain or cut items were made by Crowe et al. based on items' relation to the scale's latent construct but did not consider items' ability to distinguish narcissistic grandiosity from self-esteem—one of the main goals of the scale. For this reason, additional scrutiny is essential before concluding that the 6-item proposal provides an acceptable proxy for the full NGS.[5]

The Present Research

We sought to refine our measure of narcissistic grandiosity to more precisely differentiate narcissistic grandiosity from normative high self-esteem. To this end, we present data from four data sets, presented in two studies. In Studies 1a, 1b, and 1c, we optimize the NGS to enhance the degree to which its scores differentiate narcissistic grandiosity from self-esteem. We also examine a revision to the scale proposed by Crowe et al. (2016). In Study 2, we assess the discriminant validity of scores on the optimized NGS relative to self-esteem scores as predictors of theoretically relevant constructs.

Studies 1a, 1b, and 1c

The purpose of Study 1a was to determine which NGS items most effectively distinguished narcissistic grandiosity from self-esteem. Study 1b was used to confirm the results of Study 1a. After items for an optimized NGS were

identified in Study 1a and confirmed in Study 1b, the results of Study 1c were used to validate the item choices. Nonnarcissism criterion variables from Studies 1a and 1b provided additional evidence about the validity of the optimized NGS.

Construct Validity

In these studies, the primary convergent measure of narcissistic grandiosity was the Leadership/Authority NPI subscale (NPI-L/A; Ackerman et al., 2011) with self-esteem partialed. NPI-L/A is the closest proxy for a nonclinical narcissistic grandiosity measure among the subscales of the NPI. It includes a large number of face-valid narcissistic grandiosity items such as "I am an extraordinary person" and "I am going to be a great person." Additionally, it is the NPI subscale with the most acceptable psychometric properties (Ackerman, Donnellan, & Robins, 2012), particularly, when the NPI is administered in its original forced-choice format (Ackerman, Donnellan, Roberts, & Fraley, 2016). Independent research also indicates that NGS scores have a closer relation with NPI-L/A scores than with scores on the other NPI subscales derived by Ackerman et al. (2011; see Miller et al., 2018).

Unfortunately, many L/A items are also among the poorest items at differentiating narcissism from self-esteem (Rosenthal & Hooley, 2010; Rosenthal, Montoya, Ridings, Rieck, & Hooley, 2011). This is potentially problematic given the goal of selecting NGS items with scores relatively unencumbered by an overlap with self-esteem. Additionally, the subscale has been criticized because "leadership," per se, which is mentioned directly or indirectly in approximately half of the NPI-L/A items, is not a part of the clinical description of narcissism (e.g., Brown et al., 2009). Because of this, zero-order correlations between NGS items and NPI-L/A are not, in themselves, a suitable selection criterion for NGS items. However, as has been done using the full NPI, self-esteem can be partialed from correlations with NPI-L/A, resulting in a "purer" narcissistic grandiosity criterion relatively unencumbered by variance from self-esteem (see Brown & Zeigler-Hill, 2004; Paulhus et al., 2004; Rosenthal & Hooley, 2010). Additionally, entitlement can be partialed to further increase the specificity of the correlations with the convergent measure. Accordingly, Studies 1a, 1b, and 1c use the Ackerman et al. (2011) NPI-L/A subscale with self-esteem partialed, and also with both self-esteem and entitlement partialed, as the key convergent variables for the narcissistic grandiosity items.

In Studies 1a and 1b, in addition to NPI-L/A, we also presented zero-order and partial correlations of NGS items with the NPI Superiority/Arrogance subscale (NPI-S/A; Emmons, 1987), and in Study 1c, we presented NGS item zero-order and partial correlations with the NPI Superiority subscale (NPI-Sup; Raskin & Terry, 1988). Like NPI-L/A, NPI-S/A and NPI-Sup include face-valid narcissistic grandiosity items (e.g., NPI-S/A, "I usually dominate any conversation;" NPI-Sup, "I know that I am good because everybody keeps telling me so"). Both NPI-S/A and NPI-Sup share only one item in common with NPI-L/A, and they do not share any items with each other. In addition to NPI-L/A, NPI-S/A, and NPI-Sup, we presented NGS item correlations and partial correlations with the full NPI to investigate the convergent validity of the NGS with a general, composite measure of narcissism.

Item-Selection Criteria

An improved set of NGS items, optimized for both convergent and discriminant validity, were selected based on the following three criteria: An effective item should produce scores that are robustly related to scores on narcissistic grandiosity criterion measures (i.e., the Ackerman et al., 2011 NPI-L/A subscale with self-esteem partialed out—criterion a); it should produce scores that are more strongly related to scores on the criterion measure than to scores on a self-esteem measure (criterion b); and most important, it should produce scores that have a relation to scores on the criterion grandiosity measure that is not contingent on shared variance with self-esteem scores (criterion c). In addition, the relation of NGS item scores to scores on the criterion measure should be independent of narcissistic entitlement (i.e., a pervasive expectation of undeserved special treatment; Campbell, Bonacci, Shelton, Exline, & Bushman, 2004).

In the second stage of analysis, we investigated the convergent and discriminant validity of scores on the optimized version of the NGS relative to the Crowe et al. (2016) 6-item proposal for the NGS. Additionally, we compared the strength of relation (zero-order correlations and partial correlations controlling for self-esteem) of a 7-item optimized NGS, the original 16-item NGS, and the 6-item proposal with nonnarcissism criterion variables theoretically related to narcissistic grandiosity. These included extraversion and (dis)agreeableness (see Paulhus, 2001, for a "minimalist" description of grandiose narcissists as disagreeable extraverts) in Study 1a and (a lack of) tendency to forgive and vengeance in Study 1b (Brown, 2003, 2004).

Method

Participants and Procedure. Participants in Study 1a were 735 students (279 men, 456 women) ranging in age from 17 to 28 years (M = 18.91, standard deviation [SD] = 1.24) and in Study 1b, were 306 students (93 men, 213 women; 17-33 years of age; M = 19.28, SD = 1.95), from a university in the Northeastern United States. Participants in Study 1c were 245 students (111 men, 133 women, 1 declined to respond; 18-26 years of age; M = 20.00, SD = 1.22), from

a university in the Midwestern United States. Participants were recruited through the psychology departments' study pools. In all three studies, participants provided informed consent, then completed the measures online as parts of larger studies. To reduce priming effects, the narcissism and self-esteem measures in Studies 1a and 1b were separated by measures of other constructs. Participants were then debriefed.

Materials

Narcissistic grandiosity. The Narcissistic Grandiosity Scale is a measure of individuals' exaggerated and idealized sense of self-importance and superiority on which participants indicate how well 16 items (e.g., "superior," "envied") describe them "in general, that is, on the average" on a 1 (*not at all*) to 7 (*extremely*) scale (Study 1a: $M = 3.50$, $SD = 1.12$, $\alpha = .95$; Study 1b: $M = 3.47$, $SD = 1.12$, $\alpha = .95$; Study 1c: $M = 3.61$, $SD = 1.03$, $\alpha = .94$). The NGS was also decomposed into both an optimized 7-item version (Study 1a: $M = 3.53$, $SD = 1.21$, $\alpha = .91$; Study 1b: $M = 3.49$, $SD = 1.23$, $\alpha = .91$; Study 1c: $M = 3.66$, $SD = 1.12$, $\alpha = .90$) and the Crowe et al. (2016) 6-item proposal (Study 1a: $M = 3.49$, $SD = 1.28$, $\alpha = .92$; Study 1b: $M = 3.48$, $SD = 1.31$, $\alpha = .93$; Study 1c: $M = 3.69$, $SD = 1.08$, $\alpha = .88$).

Narcissism. The Narcissistic Personality Inventory (Raskin & Hall, 1979, 1981) is a forced-choice composite measure of narcissism as a dimensional personality trait. It includes items ranging from more normative statements such as "I am assertive" to more clearly narcissistic statements such as "I insist upon getting the respect that is due me" (see Rosenthal et al., 2011; Rosenthal & Hooley, 2010). Participants in Studies 1a and 1b completed the 37-item version of the NPI (Emmons, 1987; $M = 13.38$, $SD = 6.14$, $\alpha = .83$; and $M = 12.53$, $SD = 6.34$, $\alpha = .85$, respectively), which was also decomposed into (a) a 10-item version of the NPI-L/A subscale (Ackerman et al., 2011; $M = 5.00$, $SD = 2.34$, $\alpha = .67$; and $M = 4.87$, $SD = 2.46$, $\alpha = .71$ respectively; one item, "if I ruled the world, it would be a much better place," was not available because it is not included in the 37-item NPI) and (b) the 11-item NPI-S/A subscale (Emmons, 1987; $M = 2.81$, $SD = 2.03$, $\alpha = .59$; and $M = 2.67$, $SD = 2.14$, $\alpha = .65$ respectively). Participants in Study 1c completed the 40-item NPI (Raskin & Terry, 1988; $M = 14.39$, $SD = 6.71$, $\alpha = .83$) which was also decomposed into (a) the full 11-item NPI-L/A subscale ($M = 4.60$, $SD = 2.88$, $\alpha = .76$) and the 5-item NPI-Sup subscale ($M = 1.90$, $SD = 1.31$, $\alpha = .48$).

Self-esteem. In Studies 1a and 1b, the Rosenberg Self-Esteem Scale (Rosenberg, 1965) assessed normative feelings of self-worth. It contains 10 items such as "On the whole, I am satisfied with myself" rated on a 1 (*strongly disagree*) to 4 (*strongly agree*) scale ($M = 3.26$, $SD = 0.59$, $\alpha = .89$; and $M = 3.30$, $SD = 0.56$, $\alpha = .87$, respectively). In Study 1c, self-esteem was measured with the Self-Liking subscale of the Self-Liking and Self-Competence Scale (SL; Tafarodi & Swann, 1995) with eight items such as "I am secure in my sense of self-worth" rated on a 1 (*strongly disagree*) to 5 (*strongly agree*) scale ($M = 3.42$, $SD = 0.84$, $\alpha = .90$).

Entitlement. The Psychological Entitlement Scale (PES; Campbell et al., 2004) is a 9-item measure that assesses beliefs that one is entitled to special treatment (e.g., "I honestly feel that I'm just more deserving than others.") on a 1 (*strong disagreement*) to 7 (*strong agreement*) scale (Study 1a: $M = 2.34$, $SD = 1.09$, $\alpha = .88$; Study 1b: $M = 2.33$, $SD = 1.03$, $\alpha = .85$; Study 1c: $M = 3.06$, $SD = 1.06$, $\alpha = .83$).

Extraversion and (dis)agreeableness (Study 1a). The tendency to be sociable and energetic was measured with the 8-item Extraversion subscale of the Big-Five Inventory (John, Donahue, & Kentle, 1991) on a 1 (*disagree strongly*) to 5 (*agree strongly*) scale ($M = 3.32$, $SD = 0.77$, $\alpha = .87$). The tendency to be antagonistic toward and suspiciousness of other people was measured with the (reverse-scored) 9-item Agreeableness subscale of the same scale ($M = 3.79$, $SD = 0.58$, $\alpha = .80$).

Forgiveness and vengeance (Study 1b). The 4-item Tendency to Forgive Scale (Brown, 2003) included items such as "I tend to get over it quickly when someone hurts my feelings" and the 20-item Vengeance Scale (Stuckless & Goranson, 1992) included items such as "it is important for me to get back at people who have hurt me," both measured on a 1 (*disagree strongly*) to 7 (*agree strongly*) scale (Tendency to Forgive, $M = 4.04$, $SD = 1.24$, $\alpha = .73$; Vengeance, $M = 2.73$, $SD = 1.09$, $\alpha = .86$).

Results

Table 1 presents correlations between NGS items and multiple indices by which each item's scores were evaluated relative to measures of narcissism and self-esteem. NGS items are listed in the order of their relation in Study 1a to the key index of construct validity, the magnitude of their partial correlation with NPI-L/A with self-esteem partialed.

Convergent and Discriminant Validity

As noted in the columns of Table 1 labeled "NPI-37" and "NPI-40," an evaluation of how strongly each item relates to narcissism (criterion a) revealed that eight items correlated with the NPI-37 at $r \geq .40$ in both Studies 1a and 1b (range rs = .40−.61): authoritative, dominant, envied,

Table 1. Correlations and Partial Correlations Between NGS Items and the NPI and Measures of Self-Esteem, Studies 1a, 1b, and 1c.

Study 1a

NGS item	NPI-37	RSES	Difference (NPI-37 − RSES)	NPI-37-partial correlation controlling for RSES	NPI-37 partial correlation controlling for RSES & PES	NPI-L/A	NPI-L/A partial correlation controlling for RSES	NPI-L/A partial correlation controlling for RSES & PES	NPI-S/A	NPI-S/A partial correlation controlling for RSES	NPI-S/A partial correlation controlling for RSES & PES
Dominant[7]	.55	.16	.43	.54	.48	.53	.51	.47	.38	.37	.34
Authoritative[7]	.49	.17	.35	.47	.42	.52	.50	.46	.33	.33	.29
Powerful[7,C]	.52	.21	.35	.50	.44	.48	.46	.40	.36	.36	.32
High-status[7,C]	.47	.21	.29	.45	.37	.42	.39	.33	.29	.29	.24
Prominent[7,C]	.45	.21	.27	.43	.36	.40	.37	.32	.28	.28	.23
Superior[7]	.43	.23	.22	.42	.32	.39	.35	.28	.27	.26	.21
Envied[7]	.43	.19	.26	.42	.33	.36	.34	.28	.25	.25	.20
Heroic	.39	.23	.18	.37	.31	.35	.32	.27	.29	.29	.25
Glorious[C]	.43	.27	.18	.40	.31	.35	.31	.24	.27	.27	.22
Acclaimed[C]	.38	.24	.15	.35	.27	.34	.31	.24	.24	.23	.18
Prestigious[C]	.36	.26	.11	.34	.25	.33	.29	.22	.20	.19	.14
Extraordinary	.40	.31	.10	.37	.29	.32	.27	.20	.25	.24	.20
Omnipotent	.34	.15	.20	.32	.24	.26	.23	.16	.23	.23	.19
Unrivalled	.29	.17	.13	.27	.20	.24	.21	.15	.18	.18	.14
Brilliant	.31	.25	.07	.29	.20	.25	.20	.12	.22	.21	.16
Perfect	.30	.26	.04	.27	.18	.24	.19	.12	.16	.15	.10
NGS version											
7-item	.59[a]	.24[b]	.41[a]	.58[a]	.51[a]	.55[a]	.53[a]	.48[a]	.39[a]	.39[a]	.35[a]
16-item	.55[b]	.29[a]	.31[b]	.53[b]	.44[b]	.48[b]	.45[b]	.38[b]	.35[b]	.35[b]	.31[b]
Crowe et al. (2016) 6-item	.51[c]	.27[a]	.27[c]	.49[c]	.40[c]	.46[b]	.42[c]	.36[b]	.32[c]	.32[c]	.27[c]

(continued)

Table 1. (continued)

Study 1b

NGS item	NPI-37	RSES	Difference (NPI-37 − RSES)	NPI-37 partial correlation controlling for RSES	NPI-37 partial correlation controlling for RSES & PES	NPI-L/A	NPI-L/A partial correlation controlling for RSES	NPI-L/A partial correlation controlling for RSES & PES	NPI-S/A	NPI-S/A partial correlation controlling for RSES	NPI-S/A partial correlation controlling for RSES & PES
Dominant[7]	.61	.28	.40	.57	.51	.58	.53	.49	.36	.34	.28
Authoritative[7]	.54	.23	.35	.51	.45	.54	.50	.46	.32	.30	.24
Powerful[7,C]	.59	.34	.30	.55	.48	.52	.47	.41	.42	.40	.34
High-status[7,C]	.49	.29	.24	.44	.33	.40	.34	.25	.29	.26	.18
Prominent[7,C]	.45	.28	.20	.41	.30	.37	.32	.22	.28	.25	.17
Superior[7]	.49	.30	.23	.44	.32	.38	.32	.23	.29	.26	.18
Envied[7]	.50	.33	.21	.45	.34	.39	.31	.23	.34	.32	.25
Heroic	.34	.29	.06	.28	.21	.29	.21	.16	.24	.22	.16
Glorious[C]	.40	.27	.15	.35	.25	.32	.25	.17	.25	.23	.15
Acclaimed[C]	.41	.28	.15	.36	.25	.33	.27	.19	.23	.21	.13
Prestigious[C]	.42	.31	.13	.36	.25	.38	.31	.23	.21	.17	.09
Extraordinary	.39	.42	−.04	.31	.24	.30	.19	.13	.24	.21	.16
Omnipotent	.23	.14	.09	.19	.12	.17	.13	.07	.14	.13	.08
Unrivalled	.37	.27	.11	.31	.21	.33	.25	.18	.23	.21	.13
Brilliant	.40	.34	.07	.34	.25	.32	.24	.17	.28	.25	.19
Perfect	.33	.35	−.02	.26	.15	.26	.17	.08	.17	.13	.05
NGS version											
7-item	.64[a]	.36[a,b]	.36[a]	.60[a]	.51[a]	.56[a]	.51[a]	.44[a]	.40[a]	.38[a]	.31[a]
16-item	.58[b]	.39[a]	.25[b]	.53[b]	.43[b]	.49[b]	.42[b]	.34[b]	.36[b]	.34[b]	.26[b]
Crowe et al. (2016) 6-item	.54[c]	.35[b]	.23[b]	.49[c]	.38[c]	.45[c]	.38[c]	.30[c]	.32[c]	.30[c]	.22[c]

(continued)

Table 1. (continued)

Study 1c

NGS item	NPI-40	SL	Difference (NPI-40 − SL)	NPI-40 partial correlation controlling for SL	NPI-40 partial correlation controlling for SL & PES	NPI-L/A	NPI-L/A partial correlation controlling for SL	NPI-L/A partial correlation controlling for SL & PES	NPI-Superiority	NPI-Superiority correlation controlling for SL	NPI-Superiority partial correlation controlling for SL & PES
Dominant[7]	.42	.21	.23	.43	.38	.43	.41	.38	.28	.21	.18
Authoritative[7]	.42	.18	.18	.45	.42	.50	.50	.48	.18	.13	.11
Powerful[7,C]	.44	.29	.17	.45	.41	.48	.47	.45	.34	.29	.27
High-status[7,C]	.37	.17	.21	.39	.33	.28	.30	.27	.33	.30	.28
Prominent[7,C]	.36	.35	.01	.35	.31	.39	.35	.32	.26	.19	.16
Superior[7]	.44	.24	.22	.46	.39	.39	.39	.35	.34	.30	.27
Envied[7]	.35	.26	.10	.35	.29	.34	.33	.29	.23	.16	.12
Heroic	.30	.19	.12	.30	.22	.29	.28	.23	.21	.14	.10
Glorious[C]	.28	.23	.05	.23	.13	.27	.21	.15	.19	.13	.08
Acclaimed[C]	.22	.26	−.04	.21	.13	.24	.21	.16	.14	.09	.06
Prestigious[C]	.34	.22	.13	.34	.29	.32	.31	.28	.22	.17	.14
Extraordinary	.31	.29	.02	.28	.24	.29	.23	.21	.28	.22	.20
Omnipotent	.26	.08	.18	.26	.17	.24	.23	.18	.12	.09	.05
Unrivalled	.19	.14	.05	.20	.13	.16	.16	.12	.11	.07	.04
Brilliant	.13	.17	−.04	.17	.10	.15	.18	.14	.14	.14	.10
Perfect	.41	.23	.20	.37	.29	.32	.25	.20	.28	.25	.21
NGS version											
7-item	.51[a]	.31[a]	.24[a]	.53[a]	.48[a]	.51[a]	.51[a]	.48[a]	.36[a]	.31[a]	.28[a]
16-item	.45[b]	.30[a]	.17[b]	.47[b]	.40[b]	.44[b]	.43[b]	.39[b]	.32[a]	.27[b]	.23[b]
Crowe et al. (2016) 6-item	.43[b]	.32[a]	.13[c]	.44[b]	.37[b]	.43[b]	.41[b]	.37[b]	.32[a]	.26[b]	.23[b]

Note. Study 1a: $N = 735$; Study 1b: $N = 306$; Study 1c: $N = 245$. [7] = Item retained in the 7-item NGS; [C] = Item recommended by Crowe et al. (2016); NPI-37 = 37-item Narcissistic Personality Inventory; NPI-40 = 40-item Narcissistic Personality Inventory; RSES = Rosenberg Self-Esteem Scale; NPI-L/A = Leadership/Authority subscale of the Narcissistic Personality Inventory (Ackerman et al., 2011); NPI-S/A = Superiority/Arrogance subscale of the NPI (Emmons, 1987); NPI-Superiority = Superiority subscale of the NPI (Raskin & Terry, 1988); PES = Psychological Entitlement Scale; SL = Self-Liking subscale of the Self-Liking and Self-Competence Scale. Values above .07 are significant ($p < .05$) in Study 1a, values above .11 are significant in Study 1b, and values above .12 are significant in Study 1c. Correlation coefficients in each column labeled with different superscript letters differ significantly in magnitude ($p < .05$), whereas coefficients labeled with the same superscript letters do not ($p \geq .05$). NGS items are listed in the order of the magnitude of their relation with the key criterion (NPI-L/A with self-esteem partialed) in Study 1a.

glorious, high-status, powerful, prominent, and superior. In Study 1c, seven of those items (excluding "glorious," but with the addition of "perfect") had the eight strongest correlations with the NPI-40 (range *r*s = .35−.44).

More important, as noted in the Table 1 columns labeled "NPI-L/A," seven of those eight items (excluding "glorious") correlated with NPI-L/A at $r > .35$ in both Studies 1a and 1b (range *r*s = .36−.58). In Study 1c, six of those eight items (excluding "glorious" and "high-status") had the strongest correlations with NPI-L/A (range *r*s = .34−.50). In addition, as noted in the columns labeled "NPI-S/A," in Study 1a, the same eight items (with the addition of "heroic" and "extraordinary") were among the top ten in correlation strength with NPI-S/A (range, *r*s = .25−.38), and all but "glorious" (with the addition of "brilliant") were among the top eight in Study 1b (range, *r*s = .28−.42). As noted in the column labeled "NPI-Superiority," in Study 1c, six of the eight identified items (excluding "glorious" and "authoritative," and with the addition of "extraordinary" and "perfect") had the eight strongest correlations with NPI-Sup (range, *r*s = .23−.34).

For the relation between each item and self-esteem (criterion b; noted in the columns labeled "RSES" and "SL"), the item "extraordinary" had the strongest correlation with RSES self-esteem in Studies 1a and 1b (*r*s = .31 and .42, respectively). "Prominent" had the strongest correlation with SL self-esteem in Study 1c ($r = .35$). Of the seven items identified above that had NPI-L/A correlations of $r > .35$ in both Studies 1a and 1b (i.e., the eight originally identified items with the exception of "glorious"), "superior" had the strongest correlation with the RSES in Study 1a ($r = .23$), "powerful" had the strongest correlation in Study 1b ($r = .34$), and, as noted above, "prominent" had the strongest correlation with SL in Study 1c.

With regard to the key criterion—identifying items with narcissistic grandiosity variance not contingent on self-esteem (criterion c; noted in the Table 1 columns labeled "NPI-L/A partial correlation controlling for RSES/SL")—the seven items with the highest item-to-NPI-L/A correlations in Studies 1a and 1b were also the only items with item-to-NPI-L/A partial correlations of $r \geq .30$ across all three samples. The same seven items also had the strongest item-to-NPI-37 partial correlations controlling for self-esteem in Studies 1a and 1b (p*r*s > .40 in both samples) and were among the top eight (with the addition of "perfect") in Study 1c (p*r*s ≥ .35). These seven items, with the addition of "heroic" and "glorious" in Study 1a and "brilliant" in Study 1b, had the strongest item-to-NPI-S/A partial correlations (p*r*s ≥ .25 in both samples). In Study 1c, five of the seven items (excluding "authoritative" and "envied") were among the seven with the strongest partial correlations (with the addition of "perfect" and "extraordinary") with NPI-Sup (p*r*s ≥ .19).

Similarly, when controlling for both self-esteem and narcissistic entitlement (noted in the columns labeled "NPI-L/A partial correlation controlling for RSES/SL & PES"), the same seven items identified by their strong item-to-NPI-L/A zero-order correlations retained the seven strongest partial correlations in Study 1a (range p*r*s = .28−.47) and had seven of the eight strongest partial correlations in both Study 1b and Study 1c (with the addition of "prestigious" in both studies; Study 1b: range p*r*s = .22−.49; Study 1c: range p*r*s = .27−.48). The seven identified items also had the strongest item-to-NPI-37 partial correlations in Studies 1a and 1b (p*r*s ≥ .30 across both samples) and seven of the nine strongest correlations (with the addition of "perfect" and "prestigious") with the NPI-40 in Study 1c (range prs = .29−.42). The strongest item-to-NPI-S/A partial correlations included the same seven items and also "heroic," "glorious," and "extraordinary" in Study 1a (p*r*s ≥ .20), and "brilliant" in Study 1b, (p*r*s ≥ .17). The seven identified items had among the 10 strongest partial correlations (with the addition of "perfect," "extraordinary," and "prestigious") with NPI-Sup in Study 1c (p*r*s ≥ .11).

Results from Studies 1a, 1b, and 1c indicate that 7 of the 16 NGS items most consistently met criteria a, b, and c for final item selection. Across the three samples, these seven items were generally (a) most closely related to narcissism, (b) more closely related to narcissism than to self-esteem, and (c) their relation to narcissism was less contingent on shared variance with self-esteem or entitlement. This was the case both when considering the key criterion measure (NPI-L/A with self-esteem partialed) as well as when considering correlations and partial correlations with the full NPI. Although results were less clear when considering zero-order and partial correlations with NPI-S/A and NPI-Sup, the same seven items consistently exhibited good convergence across the three samples, whereas other items were inconsistent—the additional items that performed well in Study 1a ("heroic," "glorious," and "extraordinary") performed less well in Study 1b, and the other item that performed well in Study 1b ("brilliant") performed less well in Study 1a. Study 1c confirmed that the seven items identified in Studies 1a and 1b consistently performed well, while a number of other items also performed well in Study 1c, but less consistently so. Accordingly, the seven most consistently performing items for an optimal version of the NGS were identified and confirmed as authoritative, dominant, envied, high-status, powerful, prominent, and superior.

Comparing Versions of the NGS

Seven-Item Versus 16-Item. The distributional properties for the optimized 7-item version of the NGS were similar to those of the 16-item version (Study 1a: skewness = −.07, SE_{skew} = .09, kurtosis = −.44, $SE_{kurtosis}$ = .18; Study 1b: skewness = −.02, SE_{skew} = .14, kurtosis = −.22, $SE_{kurtosis}$ = .28; Study 1c: skewness = −.12, SE_{skew} = .16, kurtosis =

Table 2. Correlations and Partial Correlations Between NGS Versions and Nonnarcissism Criterion Variables, Studies 1a and 1b.

Study 1a					Study 1b			
NGS version	Extraversion	Extraversion partial correlation controlling for RSES	Agreeableness	Agreeableness partial correlation controlling for RSES	Tendency to Forgive (TTF)	TTF partial correlation controlling for RSES	Vengeance	Vengeance partial correlation controlling for RSES
7-item	.32[a]	.28[a]	−.08[a]	−.14[a]	−.02[a]	−.14[a]	.21[a]	.29[a]
16-item	.27[b]	.22[b]	−.03[b]	−.10[b]	.03[b]	−.10[b]	.18[a]	.26[a]
Crowe et al. (2016) 6-item	.27[b]	.22[b]	−.01[b]	−.07[c]	.03[b]	−.09[b]	.13[b]	.20[b]

Note. Study 1a: $N = 735$; Study 1b: $N = 306$. RSES = Rosenberg Self-Esteem Scale. Values above .07 are significant ($p < .05$) in Study 1a and values above .11 are significant in Study 1b. Correlation coefficients in each column labeled with different superscript letters differ significantly in magnitude ($p < .05$), whereas coefficients labeled with the same superscript letters do not ($p \geq .05$).

−.08, $SE_{kurtosis} = .31$). Table 1 (second and third rows from the bottom) reports the comparison of the 7-item NGS with the 16-item version in relation to the NPI-37 and NPI-40, the RSES and SL, NPI-LA, NPI-S/A, and NPI-Sup through the same correlations and partial correlations with which the individual items were examined (see Supplementary Material, Table S3, available online, for additional, detailed results of significance testing). Using the formula for comparing correlated correlation coefficients (Meng, Rosenthal, & Rubin, 1992), across the three studies the relation of scores on the 7-item NGS were stronger than the relation of scores on the 16-item version to NPI-L/A, the NPI-37 (Studies 1a and 1b) and NPI-40 (Study 1c), and NPI-S/A (Studies 1a and 1b), but not to NPI-Sup (Study 1c). Scores on the 7-item NGS were also correlated less strongly than were scores on the 16-item NGS with the RSES in Study 1a, but not in Study 1b, and not with the SL in Study 1c.[6]

Controlling for self-esteem, across the three studies, scores on the 7-item NGS were more strongly related than were scores on the 16-item version to NPI-L/A, the NPI-37 (Studies 1a and 1b) and NPI-40 (Study 1c), NPI-S/A (Studies 1a and 1b), and NPI-Sup (Study 1c). Represented as difference scores, scores on the 7-item NGS were more strongly related to the NPI – self-esteem correlation difference than were scores on the 16-item NGS (RSES scores measured self-esteem in Studies 1a and 1b and SL scores measured self-esteem in Study 1c).

Controlling for both self-esteem and entitlement, scores on the 7-item NGS were more strongly related than were 16-item NGS scores to NPI-L/A, the NPI-37 (Studies 1a and 1b) and NPI-40 (Study 1c), NPI-S/A (Studies 1a and 1b), and NPI-Sup (Study 1c). Taken together, scores on the 7-item version of the NGS appeared to function better than did scores on the 16-item NGS in their convergence with measures of narcissism and their ability to distinguish between narcissism and self-esteem.

Correlations and partial correlations controlling for self-esteem with the nonnarcissism criterion variables are reported in Table 2 (and detailed results of the significance tests are reported in Table S3 available online). In Study 1a, the 7-item version of the NGS was more strongly correlated with both extraversion and (dis)agreeableness than was the 16-item version, both as zero-order and partial correlations. In Study 1b, zero-order correlations with tendency to forgive were not significantly different from zero, although the difference between the 7-item and 16-item correlations with tendency to forgive was significant. Additionally, the partial correlation of the 7-item version with tendency to forgive, controlling for self-esteem, was both significantly different from zero and significantly stronger than was the partial correlation with the 16-item version. The strength of zero-order and partial correlations of scores on the 7-item and 16-item versions with vengeance did not differ.

Seven-Item Versus 6-Item. Table 1, bottom row, presents zero-order and partial correlations of the Crowe et al. (2016) 6-item proposal for the NGS with the NPI-37 and NPI-40, the RSES and SL, NPI-LA, NPI-S/A, and NPI-Sup. Descriptive statistics for the 6-item proposal were similar to those of the 7-item NGS and scale scores were similarly reliable. However, the 7-item NGS was generally more strongly related than the 6-item proposal to the convergent narcissism measures—NPI-L/A (all three studies), the NPI-37 (Studies 1a and 1b) and the NPI-40 (Study 1c), as well as NPI-S/A (Studies 1a and 1b), but not NPI-Sup (Study 1c). The 7-item NGS was also less strongly correlated with self-esteem than was Crowe et al. 6-item proposal in Study 1a, but not in Studies 1b or 1c.

Most important, the 7-item NGS performed better than the Crowe et al. (2016) 6-item proposal at distinguishing narcissistic grandiosity from self-esteem. Across the three studies, controlling for self-esteem (the key criterion), the 7-item NGS had a significantly stronger relation than the

Crowe et al. 6-item proposal with NPI-L/A. The same was true in relation to the NPI-37 (Studies 1a and 1b) and NPI-40 (Study 1c), NPI-S/A (Studies 1a and 1b), and NPI-Sup (Study 1c). Represented as difference scores, scores on the 7-item NGS were more strongly related to the NPI – self-esteem correlation difference than were scores on the 6-item proposal in all three studies (RSES scores measured self-esteem in Studies 1a and 1b and SL scores measured self-esteem in Study 1c).

Results were similar when controlling for both self-esteem and entitlement. Across the three studies, scores on the 7-item NGS retained a stronger partial correlation than did scores on the 6-item proposal with NPI-L/A, the NPI-37 (Studies 1a and 1b) and NPI-40 (Study 1c), NPI-S/A (Studies 1a and 1b), and NPI-Sup (Study 1c).

With regard to the nonnarcissism criterion variables (see Table 2 and Table S3), in Study 1a, scores on the 7-item NGS were more strongly correlated with both extraversion and (dis)agreeableness than were scores on the 6-item proposal, both as zero-order correlations and as partial correlations controlling for self-esteem. In Study 1b, zero-order correlations with tendency to forgive were not significantly different from zero, although the correlation of the 7-item NGS with tendency to forgive was significantly stronger than was the correlation of the 6-item proposal. The partial correlation controlling for self-esteem of the 7-item NGS with tendency to forgive was both significantly different from zero and significantly stronger than the partial correlation of the 6-item proposal. The 7-item version was also more strongly correlated with vengeance than was the 6-item proposal, both as zero-order correlations and as partial correlations controlling for self-esteem.[7]

Discussion

In Study 1a, we refined the NGS to improve its ability to measure narcissistic grandiosity and to distinguish it from self-esteem. We arrived at a 7-item scale that largely functioned better than the original 16-item scale but was less than half the length. We confirmed the decision to retain those seven items in Study 1b. Afterward, we validated the decision to retain the seven items with nonnarcissism criterion variables in Studies 1a and 1b. In Study 1c, we confirmed that the relations of NGS items to the key selection criterion (i.e., NPI-L/A controlling for self-esteem) were largely consistent across another independent data set.

Additionally, scores on the 7-item NGS performed better at distinguishing narcissistic grandiosity from normative self-esteem than did scores on the Crowe et al. (2016) 6-item proposal. As determined in the three samples of Study 1, when both convergent and discriminant validity are used as criteria for item selection, a different set of preferred items emerges. In all, the 7-item version of the NGS did not include three of the six items suggested by Crowe et al. (i.e., "acclaimed," "glorious," and "prestigious"). Scores on two of these items, "acclaimed" and "prestigious," discriminated relatively poorly between narcissistic grandiosity and self-esteem. Conversely, four items from the 7-item NGS were not retained by Crowe et al. ("authoritative," "dominant," "envied," and "superior"). Across the three studies, scores on "authoritative" and "dominant" functioned particularly strongly at distinguishing narcissistic grandiosity from self-esteem; they appear to be important items to retain. Taken together, scores on the 7-item NGS likely distinguish better between narcissistic grandiosity and self-esteem because some of the items provide "purer" measures of narcissistic grandiosity, relative to self-esteem, than do some of the items proposed by Crowe et al.

The 7-item version also generally outperformed the 16-item version, and more so the 6-item proposal, in relation to a number of nonnarcissism criterion measures, including extraversion, (dis)agreeableness, (lack of) forgiveness, and vengeance. In a number of cases, the relation of the NGS to the criterion variable increased in strength when self-esteem was controlled using a partial correlation, indicating the presence of a suppressor effect. However, suppressor effects when measures of narcissism and self-esteem are used as simultaneous predictors are not unusual, and may be considered valid and replicable (Paulhus et al., 2004).

Study 2

In Study 2, we investigated the convergent, criterion, and discriminant validity of scores on the optimized 7-item version of the NGS by pitting the scale head-to-head with a measure of self-esteem to determine whether NGS scores (a) outperform RSES scores when predicting constructs theoretically related more closely to narcissistic grandiosity than to self-esteem, and (b) help reveal the specificity of RSES scores in predicting constructs theoretically related more closely to self-esteem than to narcissistic grandiosity.

Study 2 thus included two types of criterion and discriminant validity measures: measures that should be more closely related to narcissistic grandiosity than to self-esteem, and measures that should be more closely related to self-esteem than to narcissistic grandiosity.

Related to Narcissistic Grandiosity

We included 10 measures theoretically linked more closely to narcissistic grandiosity than to self-esteem. They included other grandiose narcissism measures (i.e., the NPI, the PES, and the grandiose subscales of the Pathological Narcissism Inventory [PNI]; Pincus et al., 2009), self-reported superiority to others, competitiveness (Luchner, Houston, Walker, & Houston, 2011), behavioral (i.e., physical and verbal)

aggression (Bushman & Baumeister, 1998; Locke, 2009), reduced normative negative self-evaluation (i.e., less shame; Gramzow & Tangney, 1992), overreporting one's height (Moskowitz, Rieger, & Seal, 2009), and overestimating one's attractiveness (Gabriel, Critelli, & Ee, 1994, Montoya, 2008).

Related to Self-Esteem

We included 12 measures theoretically linked more closely to self-esteem than to narcissistic grandiosity. They included other self-esteem measures (Robins, Hendin, & Trzesniewski, 2001; Tafarodi & Swann, 1995), measures theoretically linked *positively* to self-esteem, including optimism (Scheier, Carver, & Bridges, 1994) and satisfaction with life (Diener, Emmons, Larsen, & Griffin, 1985; Lafrenière, Vallerand, & Sedikides, 2013), and measures theoretically linked *negatively* to self-esteem, including signs and symptoms of depression (Krizan & Herlache, 2017; Orth, Robins, Trzesniewski, Maes, & Schmitt, 2009), cognitive and affective aggression (i.e., hostility and anger; Donnellan et al., 2005; Krizan & Johar, 2015), self-reported superiority of other people compared with the self, self-reported worthlessness compared with other people (Orth et al.), seeing other people as worthless (i.e., finding flaws in others, which is linked to the nongrandiose need for rivalry rather than to the grandiose narcissistic need for admiration; Back et al., 2013; Krizan & Herlache, 2017), and manifestations of vulnerable narcissism (see Miller et al., 2017).

Method

Participants and Procedure. Participants were 147 students (71 men, 76 women) at a university in the Midwestern United States, ranging in age from 18 to 23 years (M = 19.01, SD = 1.03). Participants were recruited through a psychology department study pool. Participants were contacted approximately 2 weeks after completing the questionnaires and were asked whether they would participate in an additional session. During that session, a research assistant measured participants' height using a tape measure and took a nonsmiling photograph of each participant that included the participant's shoulders and face. At a later date, three graduate students independently rated the physical attractiveness of each participant using an unanchored 1-to-10 scale, and the ratings were averaged to form an index (M = 6.06, SD = 1.06, α_{raters} = .65).

Materials

Measures described in Study 1. Participants completed a number of measures described in Studies 1a, 1b, and 1c including the 7-item version of the NGS (M = 3.72, SD = 1.02, α = .85), the RSES (M = 3.14, SD = 0.50, α = .88), the NPI-40 (M = 15.68, SD = 6.25, α = .78), and the PES (M = 3.11, SD = 0.98, α = .84).

Pathological narcissism. The PNI (Pincus et al., 2009) is a 52-item self-report measure of narcissism with subscales that assess characteristics of grandiose and vulnerable narcissism on a 0 (*not at all like me*) to 5 (*very much like me*) scale. As in Crowe et al. (2016), the three grandiose subscales were averaged to form an index of grandiose narcissism (M = 2.88, SD = 0.61, α = .80), and the four vulnerable subscales were averaged to form an index of vulnerable narcissism (M = 2.20, SD = 0.72, α = .93).

Self-esteem. The Self-Liking and Self-Competence Scale (SLSC; Tafarodi & Swann, 1995) and the Single-Item Self-Esteem Scale (SISE; Robins et al., 2001) assess the normative range of self-evaluations. For the SLSC, participants rate affective self-regard (self-liking; M = 3.63, SD = 0.81, α = .91) and beliefs about their abilities (self-competence; M = 3.32, SD = 0.57, α = .79), using 8-item subscales with items rated on a 1 (*strongly disagree*) to 5 (*strongly agree*) scale. For the SISE, respondents indicate whether they have high self-esteem on a 1 (*not very true of me*) to 5 (*very true of me*) scale (M = 3.38, SD = 1.07).

Competitiveness. The competitiveness subscale of Wagner's (1995) individualism and collectivism measure includes five items such as "doing your best isn't enough; it is important to win" measured on a 1 (*strongly disagree*) to 7 (*strongly agree*) scale (M = 3.37, SD = 1.17, α = .80).

Superiority and worthlessness. Four items developed by the authors assessed four aspects of individuals' superiority and worthlessness beliefs (i.e., there was one item per assessed belief): "compared to most other people, I am superior" (M = 3.16, SD = 1.57), "compared to most other people, I am worthless" (M = 1.91, SD = 1.20), "compared to me, most other people are superior" (M = 2.92, SD = 1.54), and "compared to me, most other people are worthless" (M = 1.81, SD = 1.07), each rated on a 1 (*not at all true*) to 7 (*very true*) scale.

Behavioral, affective, and cognitive aggression. Behavioral, affective, and cognitive manifestations of aggression were assessed with the Aggression Questionnaire (Buss & Perry, 1992), with items measured on a 1 (*extremely uncharacteristic of me*) to 7 (*extremely characteristic of me*) scale. Behavioral aggression was assessed with nine physical aggression items such as "once in a while I can't control the urge to strike another person" (M = 2.90, SD = 1.24, α = .85), and five verbal aggression items such as "I can't help getting into arguments when people disagree with me" (M = 3.78, SD = 1.24, α = .80). Cognitive aggression (i.e., hostility) was assessed with eight items such as "I am suspicious of overly friendly strangers" (M = 3.32, SD = 1.53, α = .86). Affective aggression (i.e., anger) was assessed with seven items such as "sometimes I fly off the handle for no good reason" (M = 3.13, SD = 1.17, α = .78).

Shame proneness. The tendency to experience normative shame when caught in a social transgression was assessed using the Negative Self-Evaluation subscale of the Guilt and Shame Proneness Scale (Cohen, Wolf, Panter, & Insko, 2011). Participants rate the likelihood of shame reactions (e.g., "What is the likelihood that this would make you feel like a bad person?") to four social transgressions on a 1 (*very unlikely*) to 7 (*very likely*) scale ($M = 5.47$, $SD = 1.02$, $\alpha = .61$).

Satisfaction with life. The 5-item Satisfaction With Life Scale (Diener et al., 1985) assesses individuals' sense of fulfillment and contentment using items such as "in most ways, my life is close to my ideal" rated on a 1 (*strongly disagree*) to 7 (*strongly agree*) scale ($M = 4.87$, $SD = 1.21$, $\alpha = .84$).

Optimism. The 9-item Life Orientation Test (modified version; Scheier et al., 1994) assesses optimism—the tendency to focus on the positive side of events or conditions and to expect favorable outcomes—with items such as "in uncertain times, I usually expect the best" rated on a 1 (*strongly disagree*) to 5 (*strongly agree*) scale ($M = 3.43$, $SD = 0.56$, $\alpha = .78$).

Depression. The Center for Epidemiologic Studies Depression Scale (CES-D; Radloff, 1977) assesses 20 signs or symptoms of depression in members of the general population on a 0 (*rarely or none of the time*) to 3 (*much or all of the time*) scale ($M = 0.88$, $SD = 0.50$, $\alpha = .90$).

Height and attractiveness. Participants reported their height in feet and inches and rated their own attractiveness on an unanchored 1-to-10 scale ($M = 5.92$, $SD = 1.30$).

Results

To assess the criterion and discriminant validity of NGS and RSES scores, we conducted relative weights analyses (RWA; Johnson, 2000; Tonidandel, LeBreton, & Johnson, 2009). We used RWA, rather than linear regressions, because RWA produces more accurate results in situations in which predictor variables have high intercorrelations (in Study 2, the correlation between the NGS and RSES was $r = .42$). Even though correlated measures of narcissism and self-esteem may produce interpretable, replicable suppressor effects when paired as predictors in a linear regression (see Paulhus et al., 2004), RWA is preferable because it treats each predictor as orthogonal to every other predictor. As a result, RWA can more accurately estimate the variance explained by the correlated predictor variables. The relative weight for each predictor is equivalent to the R^2 of that predictor, and therefore the weights in each equation sum to the total R^2 for the equation. The percentage of the total R^2 accounted for by each predictor relative to the other predictors can be calculated by dividing each weight by the sum of all the weights.

A bootstrapping method developed by Tonidandel et al. (2009) generates a 95% confidence interval (CI) for each weight. The CI determines (a) whether each predictor accounts for a significant amount of variance in the dependent variable and (b) whether the variance accounted for by a predictor is significantly different from the variance accounted for by the other predictor(s). Zero-order correlations, the total R^2 for each model, unstandardized predictor variable weights (i.e., the R^2 for each predictor) and confidence intervals, and the percentage of the R^2 accounted for by each predictor are reported in Tables 3 and 4. We expected that the NGS would predict a higher percentage than would the RSES for each outcome variable theoretically linked more closely to narcissistic grandiosity than to self-esteem, and that the RSES would predict a higher percentage than would the NGS for each outcome variable linked more closely to self-esteem than to narcissistic grandiosity. Because the goal of the study was to understand whether NGS scores were consistently differentiable from RSES scores in relation to other theoretically relevant constructs, rather than to identify the precise relation of narcissistic grandiosity and self-esteem to each outcome variable, we did not adjust the p values to account for multiple comparisons.

Constructs Linked to Narcissistic Grandiosity

Table 3 reports the results of the RWA analysis using narcissistic grandiosity (i.e., the NGS) and self-esteem (i.e., the RSES) as simultaneous predictors of constructs theoretically linked more strongly to narcissistic grandiosity than to self-esteem. First, narcissistic grandiosity was more strongly related to other measures of grandiose narcissism than was self-esteem. For instance, for the NPI, narcissistic grandiosity explained 86.7%, compared with 13.3% explained by self-esteem, of the total R^2 of .30. Similarly, grandiosity explained 91.4%, compared with 8.6% explained by self-esteem, of the total R^2 of .13 for entitlement, 85.9%, compared with 14.1%, of the total R^2 of .07 for the PNI grandiose index, and 87.5%, compared with 12.5%, of the total R^2 of .24 for self-reported superiority relative to others.

Second, grandiosity explained higher percentages of the R^2 than did self-esteem for other constructs theoretically linked to narcissistic grandiosity. These included competitiveness (90.6% vs. 9.4% of the total R^2 of .12), behavioral aggression (i.e., physical aggression, 79.5% vs. 20.5% of the total R^2 of .09; verbal aggression, 86.8% vs. 13.2% of the total R^2 of .05), and (a lack of) shame (93.0% vs. 7.0% of the total R^2 of .15).

Third, we examined the relation of narcissistic grandiosity and self-esteem to two observation-verified self-reported

Table 3. Correlations and Relative Weights Analysis Predicting Constructs Theoretically Linked to Narcissistic Grandiosity, Study 2.

Outcome	Predictor	*r*	Relative weight/R^2_{IV} (95% CI)	Percent R^2 explained
Narcissistic Personality Inventory-40 (R^2 = .30)	NGS	.55*	.26* (.16, .38)[a]	86.7
	RSES	.28*	.04 (.01, .10)	13.3
Entitlement (R^2 = .13)	NGS	.32*	.11* (.03, .23)[a]	91.4
	RSES	.00	.01 (.00, .04)	8.6
Pathological Narcissism Inventory: Grandiosity (R^2 = .07)	NGS	.22*	.06* (.01, .16)[a]	85.9
	RSES	−.03	.01 (.00, .05)	14.1
Superiority compared with others (R^2 = .24)	NGS	.49*	.21* (.09, .34)[a]	87.5
	RSES	.24*	.03 (.01, .09)	12.5
Competitiveness (R^2 = .12)	NGS	.32*	.11* (.03, .24)[a]	90.6
	RSES	.01	.01 (.00, .04)	9.4
Physical aggression (R^2 = .09)	NGS	.23*	.08* (.01, .15)[a]	79.5
	RSES	−.06	.01 (.00, .07)	20.5
Verbal aggression (R^2 = .05)	NGS	.19*	.04* (.01, .13)[a]	86.8
	RSES	−.02	.01 (.00, .04)	13.2
Shame (R^2 = .15)	NGS	−.39*	.14* (.06, .25)[a]	93.0
	RSES	−.15	.01 (.00, .04)	7.0
Self-reported height (R^2 = .88)	Actual height	.93*	.84* (.63, .93)	96.2
	NGS	.25*	.03* (.01, .08)[a]	3.6
	RSES	.05	.00 (.00, .00)	0.2
Self-assessed attractiveness (R^2 = .19)	Rater-assessed attractiveness	−.02	.00 (.00, .00)	0.3
	NGS	.36*	.12* (.00, .23)	61.5
	RSES	.35*	.07 (.00, .16)	38.2

Note. N = 147. NGS = Seven-Item Narcissistic Grandiosity Scale; RSES = Rosenberg Self-Esteem Scale; R^2_{IV} = percentage of variance in the predicted variable accounted for by the NGS or RSES score; [a]NGS contributes more relative variance than does the RSES to the prediction of the outcome.
*Relation with the predicted variable was significant at $p < .05$.

constructs theoretically linked to narcissistic grandiosity: overreporting one's height and overestimating one's attractiveness. For height, individuals' actual height explained 96.2% of the total R^2 of .88. However, narcissistic grandiosity contributed significantly more to self-reports of height (3.6%) than did self-esteem (0.2%), which was not a significant predictor. In other words, narcissistic grandiosity, but not self-esteem, predicted exaggeration of an individual's height beyond their actual height. For attractiveness (total R^2 = .19), observer ratings were unrelated to self-reports (observer ratings explained 0.3% of the R^2). In this case, the difference between the percentage of participants' attractiveness self-assessments explained by narcissistic grandiosity (61.5%) and by self-esteem (38.2%) did not differ significantly. However, grandiosity was a significant predictor of self-assessed attractiveness whereas self-esteem was not.

Constructs Linked to Self-Esteem

Table 4 reports the results of the RWA analysis using narcissistic grandiosity (i.e., the NGS) and self-esteem (i.e., the RSES) as simultaneous predictors of constructs theoretically linked more strongly to self-esteem (either positively or negatively) than to grandiosity. First, the RSES was more strongly related to other measures of self-esteem than was grandiosity. The RSES explained 82.5%, compared with 17.5% explained by narcissistic grandiosity, of the of the total R^2 of .76 for self-liking, and 76.8%, compared with 23.2% explained by narcissistic grandiosity, of the total R^2 of .54 for self-competence. The RSES also explained 65.7%, compared with 34.3% explained by narcissistic grandiosity, of the total R^2 of .54 for the single-item measure of self-esteem.

Second, self-esteem explained higher percentages of the R^2 than did narcissistic grandiosity for other (non-self-esteem) constructs theoretically linked to *higher* levels of self-esteem. These included optimism (87.4% vs. 12.6% of the total R^2 of .27) and satisfaction with life (85.4% vs. 14.6% of the total R^2 of .40).

Third, self-esteem explained higher percentages of the R^2 for other constructs theoretically linked to *lower* levels of self-esteem than did grandiosity. These included signs and symptoms of depression (87.1% vs. 12.9% of the total R^2 of .45), cognitive aggression (i.e., hostility; 93.0% vs. 7.0% of the total R^2 of .23), affective aggression (i.e., anger, 90.9% vs. 9.1% of the total R^2 of .06), judging others to be superior to oneself (94.8% vs. 5.2% of the total R^2 of .14), judging oneself to be worthless compared with others (79.9% vs. 20.1% of the total R^2 of .46), and disparaging others as worthless compared with oneself (92.2% vs. 7.8% of the total R^2 of .11). Self-esteem also explained a higher percentage of the R^2 than did grandiosity for the PNI vulnerable narcissism index (95.3% vs. 4.7% of the total R^2 of .19).

Table 4. Correlations and Relative Weights Analysis Predicting Constructs Theoretically Linked to Self-Esteem, Study 2.

Outcome	Predictor	*r*	Relative weight/R^2_{IV} (95% CI)	Percent R^2 explained
Self-liking (R^2 = .76)	NGS	.50*	.13* (.07, .20)	17.5
	RSES	.86*	.63* (.54, .70)[a]	82.5
Self-competence (R^2 = .54)	NGS	.47*	.13* (.04, .22)	23.2
	RSES	.72*	.42* (.30, .53)[a]	76.8
Single-item self-esteem (R^2 = .54)	NGS	.55*	.19* (.10, .27)	34.3
	RSES	.68*	.36* (.24, .47)[a]	65.7
Optimism (R^2 = .27)	NGS	.26*	.03 (.01, .10)	12.6
	RSES	.52*	.23* (.10, .37)[a]	87.4
Satisfaction with life (R^2 = .40)	NGS	.33*	.06* (.02, .12)	14.6
	RSES	.63*	.34* (.24, .45)[a]	85.4
Depression (R^2 = .45)	NGS	−.33*	.06* (.01, .13)	12.9
	RSES	−.66*	.39* (.26, .51)[a]	87.1
Hostility (R^2 = .23)	NGS	−.17*	.02 (.00, .04)	7.0
	RSES	−.48*	.21* (.11, .34)[a]	93.0
Anger (R^2 = .06)	NGS	.01	.01 (.00, .03)	9.1
	RSES	−.23*	.06* (.03, .15)[a]	90.9
Viewing others as superior (R^2 = .14)	NGS	−.09	.01 (.00, .03)	5.2
	RSES	−.37*	.13* (.04, .24)[a]	94.8
Worthlessness compared with others (R^2 = .46)	NGS	−.41*	.09* (.04, .19)	20.1
	RSES	−.67*	.37* (.23, .49)[a]	79.9
Viewing others as worthless (R^2 = .11)	NGS	−.02	.01 (.00, .03)	7.8
	RSES	−.31*	.10* (.02, .24)[a]	92.2
Pathological Narcissism Inventory: Vulnerability (R^2 = .19)	NGS	−.10	.01 (.00, .03)	4.7
	RSES	−.43*	.18* (.08, .30)[a]	95.3

Note. N = 147. NGS = Seven-Item Narcissistic Grandiosity Scale; RSES = Rosenberg Self-Esteem Scale. R^2_{IV} = percentage of variance in the predicted variable accounted for by the NGS or RSES score; [a]RSES contributes more relative variance than does the NGS to the prediction of the outcome.
*Relation with the predicted variable was significant at $p < .05$.

Discussion

Study 2 provides evidence (a) of convergent and discriminant validity for both NGS and the RSES scores, particularly when the two measures are used in tandem. The NGS outperformed the RSES in predicting other measures of grandiose narcissism, whereas the RSES outperformed the NGS in predicting other measures of self-esteem. The results also provide evidence (b) of NGS scores' concurrent validity based on the strength of their relation, relative to RSES scores, to constructs theoretically linked more closely to narcissistic grandiosity than to self-esteem. For instance, the NGS helps clarify that narcissistic grandiosity is more closely linked than is self-esteem to competitiveness (e.g., Luchner et al., 2011) and aggressive behavior (i.e., physical and verbal outbursts; Bushman & Baumeister, 1998). Narcissistic grandiosity is also linked to self-reported "shamelessness," rather than to heightened feelings of shame, as posited by some long-held theories (e.g., Kernberg, 1975; Kohut, 1977). There is also preliminary evidence that narcissistic grandiosity is more closely linked than self-esteem to exaggerated self-reports of height (Moskowitz et al., 2009), and overestimation of one's attractiveness (Gabriel et al., 1994).

Additionally, the results provide evidence (c) of concurrent criterion-related validity for RSES scores based on the strength of their relation, relative to NGS scores, to constructs theoretically linked more closely to self-esteem than to narcissistic grandiosity. In contrast to the findings of Baumeister et al. (2003), higher self-esteem, when factored alongside narcissistic grandiosity, is related to self-reports of optimism (Scheier et al., 1994) and life satisfaction (Diener et al., 1985). Similarly, lower self-esteem is related to depression (Krizan & Herlache, 2017; Orth et al., 2009), feelings of worthlessness compared with others (Orth et al., 2009), and beliefs that others are both superior (Orth et al., 2009) and worthless (Krizan & Herlache, 2017). Lower self-esteem is also related to vulnerable narcissism (see Miller, Lynam, Vize, et al., 2017), whereas the NGS is not. This suggests that the weak empirical relation between grandiose and vulnerable narcissism continues to warrant discussion (e.g., Krizan & Herlache, 2017; Miller, Lynam, Hyatt, & Campbell, 2017).

The findings are particularly noteworthy in a number of cases in which both the NGS and RSES have significant zero-order relations with the predicted variables. For instance, although both grandiosity and self-esteem are positively correlated with self-assessed superiority compared with others, only grandiosity's weight was significant in the RWA, indicating that grandiosity is the more robust predictor. Similarly, although both self-esteem and grandiosity are positively correlated with

optimism, only self-esteem's weight was significant, indicating that self-esteem is the more robust predictor (Brown et al., 2009, identified a similar pattern using the NGS and RSES as simultaneous predictors of mental health).

Using the NGS and RSES together as predictors may also help clarify the debate over the relation (or lack thereof) of self-esteem to aggression. As Baumeister et al. (2003) noted, high self-esteem scores are typically "associated with both the presence and absence of aggression" (p. 6). As with Donnellan et al. (2005; Study 3), we found that narcissistic grandiosity was more closely related to behavioral (i.e., physical and verbal) aggression, whereas low self-esteem was more closely related to cognitive and affective aggression (i.e., hostility and anger). Research that examines behavioral, cognitive, and affective aggression separately may help parse the seemingly conflicting conclusions reached by different researchers (e.g., Baumeister et al., 2003; Donnellan et al., 2005; Krizan & Johar, 2015) regarding the relation of self-esteem to aggression.

In summary, Study 2 provides consistent evidence that the NGS and RSES are distinct in their relations to theoretically relevant constructs. It also provides evidence that narcissistic grandiosity is not at the top of a continuum of self-report self-esteem (as suggested by Baumeister et al., 2003), but rather, is an independent, measurable, and differentiable construct.

General Discussion

Using four data sets across two studies, we provided evidence for a refined and improved version of the NGS. In Studies 1a and 1b, the relations of each NGS item to measures of narcissism and self-esteem were assessed to develop an optimized 7-item NGS that both measured the target construct and provided optimal discriminant validity from the effects of self-esteem (as well as narcissistic entitlement). The results of Study 1c validated the choices of items for the optimized NGS. Study 2 indicated that the NGS, in combination with the RSES, differentially predicted constructs theoretically relevant to scores on each measure: NGS scores were more strongly related to verbal and physical aggression, competitiveness, and shamelessness whereas RSES scores were more strongly related positively to optimism and satisfaction with life and negatively to depression, worthlessness, hostility, and anger.

Narcissistic Grandiosity Versus High Self-Esteem

We developed the NGS not only to introduce a measure of narcissistic grandiosity to the literature but also to supplement the literature on self-esteem with a scale that can help tease apart the difference between normative positive self-regard (i.e., high self-esteem) and an inflated sense of self (i.e., narcissistic grandiosity). Narcissistic grandiosity and high self-esteem overlap semantically, and their self-reports are positively correlated. However, mounting evidence suggests that they are distinct from each other (e.g., Brummelman et al., 2016), rather than a unitary, continuous construct with narcissistic grandiosity at the top (e.g., Baumeister et al., 2003). Regarding the latter point of view, Swann, Chang-Schneider, and McClarty (2007) pointedly noted that "conflating narcissism and true high self-esteem is profoundly problematic for the same reason that it is problematic to mistake for a friend an enemy who is merely masquerading as a friend" (p. 87).

Viewing narcissistic grandiosity as lying at the top of the self-esteem continuum has potential real-world consequences. For instance, the work of Baumeister et al. (2003) described earlier was widely reported in the popular media. Although national headlines such as "The Trouble With Self-Esteem" (Slater, 2002) and "Lacking in Self-Esteem? Good for You!" (Sullivan, 2002), may have been sensationalistic, they conveyed a public message that has carried forward for more than a decade (e.g., Singal, 2017) that "too much" self-esteem can be detrimental, so one is better off with less of it. Our results indicate that when narcissistic grandiosity is (properly, in our view) differentiated from high self-esteem, self-esteem's relation to constructs that are crucial to individuals' well-being, such as optimism, satisfaction with life, depression, worthlessness, hostility, and anger, can be more clearly and meaningfully evaluated.

Which Version of the NGS Should Researchers Use?

Numerous peer-reviewed studies have already used the 16-item NGS. Additionally, we proposed and tested a 7-item version of the scale (see the Supplementary Material, Appendix S1, for both versions). Scores on both versions appear to measure the target construct well and to differentiate it from high self-esteem. However, we do not recommend limiting the NGS to the six items proposed by Crowe et al. (2016). Based on the results of Studies 1a, 1b, and 1c, scores on a number of the items they suggest retaining (i.e., "acclaimed" and "prestigious") discriminate relatively poorly between narcissism and self-esteem, whereas scores on a number of the items they suggest dropping (i.e., "authoritative" and "dominant") appear to discriminate particularly well. Our results suggest that, in aggregate, the Crowe et al. items measure a version of narcissistic grandiosity that differentiates less well between narcissistic grandiosity and self-esteem than do scores on the 7-item NGS. It is important for researchers who may suggest revisions to the NGS to consider its theoretical foundations to guide their work.

Limitations

There are a number of limitations to the current research. First, the convergent narcissistic grandiosity variables for

evaluating NGS items and recommending a shortened version of the scale were limited to the NPI and its most relevant subscales. Results using two of those subscales—NPI-S/A and NPI-Sup—were less clear-cut than were results using the full NPI and NPI-L/A, although this may be attributable in part to the poor internal consistency of the forced-choice versions of NPI-S/A and NPI-Sup. Other convergent measures specific to narcissistic grandiosity (e.g., the Grandiosity subscale of the Narcissistic Admiration and Rivalry Questionnaire, Back et al., 2013), or even a Likert-type version of the NPI (see Ackerman et al., 2016; Grosz et al., 2019) might reveal that different NGS items are preferable, or conversely, might strengthen confidence in the seven selected items. However, even given this limitation, our theoretical focus on discriminant validity and our results comparing the 7- and 16-item versions of the NGS with the Crowe et al. (2016) 6-item proposal mitigate against use of the Crowe et al. 6-item proposal without additional research.

Second, it would be beneficial to examine more nonnarcissism criterion variables, and to have additional independent samples, to test the effectiveness of different versions of the NGS. However, results using Big-Five extraversion and agreeableness (Study 1a) and tendency to forgive and vengeance (Study 1b) provide initial evidence that scores on the 7-item NGS provide a greater level of predictive specificity than do scores on the 16-item version, and to a greater extent, the 6-item proposal. It would also be valuable to conduct experimental tests with behavioral dependent variables to help determine whether the differences between NGS versions, and differences in the ability of the NGS versus measures of self-esteem to predict theoretically relevant outcomes, are ecologically meaningful. Additionally, the reported studies relied on college/university students as participants. Results may differ among more diverse samples.

Conclusion

The NGS is a short measure of narcissistic grandiosity with scores that exhibit good reliability and validity. Its scores represent the target construct, narcissistic grandiosity, well, and help differentiate it from its semantic near-neighbor, high self-esteem. The scale has already been used in numerous independent peer-reviewed studies in multiple research labs both in and outside the United States (see Table S2). As a measure of a specific facet of grandiose narcissism, the NGS can be (and has already been) combined with measures of other narcissism facets to paint a more precise and nuanced picture of narcissism than is possible with multifaceted composite narcissism scales such as the NPI (e.g., Brown et al., 2009; Brunell & Buelow, 2018; Buelow & Brunell, 2014; Luo et al., 2014; Tamborski et al., 2012). Evidence also indicates that the NGS can be used in tandem with a measure of self-esteem to help clarify the important differences between the two constructs, both in theory, and as predictors of important real-life characteristics such as optimism, life satisfaction, worthlessness, depression, and different forms of aggression. We hope that the NGS continues to help spur a better understanding of both narcissism and self-esteem and their relation to one another.

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Notes

1. Narcissistic grandiosity also differs from grandiose narcissism. Narcissistic grandiosity—an exaggerated and idealized sense of self-importance and superiority—is a narrowly defined facet of grandiose narcissism. Grandiose narcissism, in contrast, is a broadly defined type of narcissism that includes not only narcissistic grandiosity but also entitled beliefs, exploitative demands, arrogant, manipulative, and hostile behaviors, lack of empathy, and so on. Additionally, grandiose narcissism contrasts with vulnerable narcissism—a narcissism subtype in which individuals are less likely to exhibit grandiose beliefs and behaviors, and instead are prone to experience shame, humiliation, low self-esteem, and depression (see, Cain, Pincus, & Ansell, 2008; Ronningstam, 2005). This article uses the generic term *narcissism* to refer specifically to grandiose narcissism except where otherwise noted.
2. This problem is most likely not confined to Rosenberg's (1965) scale and may affect most or all current self-report measures of self-esteem. In fact, Rosenberg's scale appears to be one of the better self-esteem measures at discriminating between normative and egotistical responding (Brown & Zeigler-Hill, 2004).
3. Since the development of the NGS, a number of measures with narcissistic grandiosity subscales have been published. Two of them (Krueger, Derringer, Markon, Watson, & Skodol, 2012; Simms et al., 2011) are not appropriate for disentangling narcissistic grandiosity from high self-esteem because they measure clinically maladaptive narcissistic grandiosity and also contain some items that appear to measure entitlement (e.g., "I deserve special treatment"). A third, the Narcissistic Admiration and Rivalry Questionnaire (Back et al., 2013), measures nonclinical, trait narcissism

(like the NGS) and includes a 3-item Grandiosity subscale. That subscale has lower internal consistency than the NGS but correlates strongly with it. Unlike the NGS, none of the three newer measures, can be easily adapted to measure state changes in narcissistic grandiosity as originally intended (Rosenthal, 2005; see also, Crowe, Edershile, et al., 2018).

4. Our own unpublished data (N = 141) also indicated that NGS scores demonstrate good 1-month test–retest reliability ($r_{test–retest}$ = .78).
5. Although we maintain that IRT is not an appropriate method for optimizing the NGS for a number of reasons, an additional concern regarding the Crowe et al. (2016) results is that it may also be problematic to make item-reduction proposals based on a sample-specific solution. To investigate this issue, we conducted an ad hoc IRT analysis using NGS data collected previously (see van der Linden & Rosenthal, 2016). We used the same methodology and sampling platform (i.e., Amazon Mechanical Turk) as Crowe et al., and our sample was larger (N = 2,145 vs. N = 870). We found that a number of the results on which Crowe et al. based their alternate proposal for the NGS did not replicate in our sample. For instance, two items dropped by Crowe et al. for poor fit ("brilliant" and "unrivaled") exhibited acceptable fit in our analysis, and two items proposed by Crowe et al. for a 6-item NGS ("acclaimed" and "prestigious") exhibited poor fit. Additional details are presented in the Supplementary Material (see Table S4).
6. In Studies 1a, 1b, and 1c, the correlations between the 16-item and 7-item NGS were r = .96, r = .95, and r = .94, respectively, the correlations between the 16-item NGS and the 6-item proposal were r = .95, r = .95, and r = .96, respectively, and the correlations between the 7-item NGS and the 6-item proposal were r = .92. r = .91, and r = .92, respectively.
7. The 16-item NGS also outperformed the 6-item proposal on many of the same comparisons (Table 1, bottom two rows, and Table S3, include summaries of significance tests).

Supplemental Material

Supplemental material for this article is available online.

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